Forest Cover Type Classification Project: A Machine Learning Approach

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Abstract

This project focuses on using machine learning to predict different types of forest cover in wilderness areas based on geographic and topographic data. The Forest Cover Type dataset from the UCI Machine Learning Repository contains over 580,000 samples representing 30×30 meter cells of forest with 54 features. Three machine learning models—Logistic Regression, Random Forest, and LightGBM—are implemented and compared. The analysis reveals that ensemble methods significantly outperform Logistic Regression, with Random Forest achieving the highest macro F1 score of 0.915 on test data. Elevation and distances to roadways and fire points are identified as the most important predictive features. Misclassification analysis shows that errors often occur at boundaries between forest types, particularly in transition zones. This work demonstrates how machine learning can help understand complex ecological systems and inform forest management decisions.

1 Introduction

Understanding forest ecosystems and their distribution is crucial for ecological management, conservation efforts, and climate change research. The composition of forest types across landscapes is influenced by various geographic and topographic factors, making it an interesting and complex prediction problem well-suited for machine learning approaches.

This project focuses on using machine learning to predict different types of forest cover in wilderness areas based on geographic and topographic data. It's a practical application of classification algorithms that demonstrates how landscape features can help us understand forest ecosystems. By accurately predicting forest cover types from terrain data, we can better manage forest resources, monitor changes in ecosystems over time, and potentially model how climate change might affect forest distributions.

The specific objectives of this project are:

1. To explore and visualize the Forest Cover Type dataset to understand the relationships between terrain features and forest types
2. To implement and compare multiple machine learning algorithms for classifying forest cover types
3. To identify the most important topographic and geographic features for forest type prediction
4. To analyze where and why misclassifications occur in the models

The classification problem addressed here is challenging for several reasons. First, the dataset exhibits significant class imbalance, with some forest types representing less than 1% of the samples. Second, forest types often exist in transition zones rather than having clear boundaries, creating inherent ambiguity in some regions. Third, the high dimensionality of the feature space (54 features) requires careful analysis to identify which factors truly drive forest type distribution.

We implemented and compared three machine learning approaches with different levels of complexity: Logistic Regression as a baseline linear model, Random Forest as a powerful ensemble method, and LightGBM with hyperparameter optimization as a state-of-the-art gradient boosting approach. This comparison allows us to evaluate the trade-offs between model simplicity, interpretability, and predictive performance.

The work demonstrates the practical application of machine learning techniques to a real-world ecological classification problem, providing insights into how different algorithms perform and which terrain features are most relevant for forest type prediction.

2 Dataset

2.1 Dataset Overview

The Forest Cover Type dataset comes from the UCI Machine Learning Repository and contains information about forest areas in the Roosevelt National Forest in Colorado. This dataset is substantial, with over 580,000 samples, each representing a 30×30 meter cell of forest.

The dataset has the following characteristics:

- It has 7 different classes of forest cover types (lodgepole pine, spruce/fir, ponderosa pine, etc.)
- It includes 54 features that describe the terrain and environment

- The features include elevation, slope, aspect (direction the slope faces), distances to water, roads, and fire points, hillshade measurements at different times of day, and binary indicators for wilderness areas and soil types
- It represents a real-world ecological classification problem with practical applications

2.2 Feature Description

The dataset contains the following types of features:

1. **Numerical Features (10):**
   - Elevation (meters)
   - Aspect (degrees azimuth)
   - Slope (degrees)
   - Horizontal distance to hydrology (meters)
   - Vertical distance to hydrology (meters)
   - Horizontal distance to roadways (meters)
   - Hillshade at 9am
   - Hillshade at noon
   - Hillshade at 3pm
   - Horizontal distance to fire points (meters)
2. **Binary Features (44):**
   - Wilderness areas (4 binary columns)
   - Soil type (40 binary columns)

2.3 Exploratory Data Analysis

2.3.1 Class Distribution

The dataset exhibits significant class imbalance, as shown in Figure 1. Classes 1 and 2 (Spruce/Fir and Lodgepole Pine) dominate the dataset, comprising approximately 85% of the samples (36.46% and 48.76% respectively), while Class 4 (Cottonwood/Willow) represents only 0.47% of the data with just 2,747 samples. Classes 5–7 each form around 1–3% of the dataset.

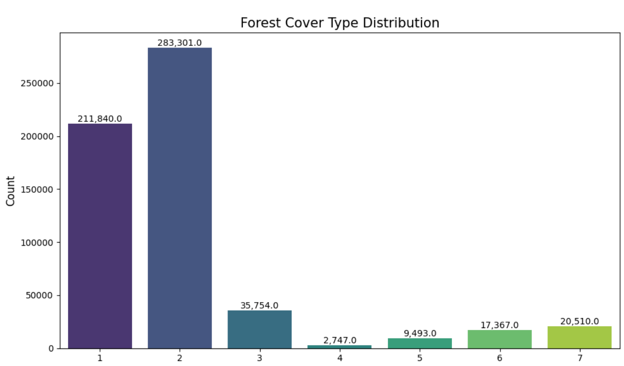

Figure 1: Distribution of Forest Cover Types

This imbalance makes classification challenging, as models can achieve high overall accuracy by mainly predicting the majority classes while struggling to correctly identify the minority classes.

2.3.2 Elevation Distribution

Elevation is strongly associated with forest cover types and varies markedly across them. The elevation range spans roughly 1,860 m to 3,858 m, with different forest types occupying distinct elevation bands.

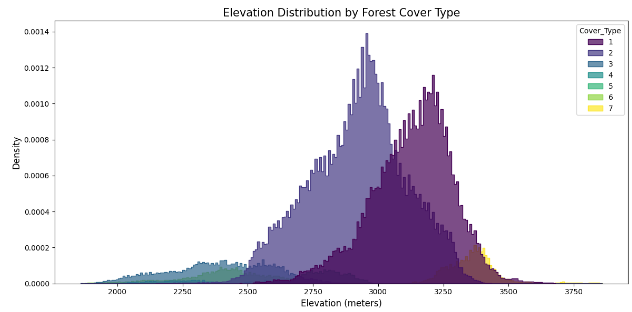

Figure 2: Elevation Distribution by Forest Cover Type

As shown in Figure 2, Class 7 (Krummholz) occurs only at the highest elevations (near tree line, above 3,300 m), whereas Class 4 (Cottonwood/Willow) is found at very low elevations in valley bottoms. Other classes occupy mid-range elevations – Class 3 (Ponderosa Pine) is generally at lower-mid elevations while Classes 1 (Spruce/Fir) and 2 (Lodgepole Pine) inhabit higher montane zones – but still show differing distribution peaks. These elevation-based separations explain why Elevation emerged as a critical predictor for cover type.

2.3.3 Feature Correlation

Analysis of correlations between numerical features reveals several important relationships (Figure 3).

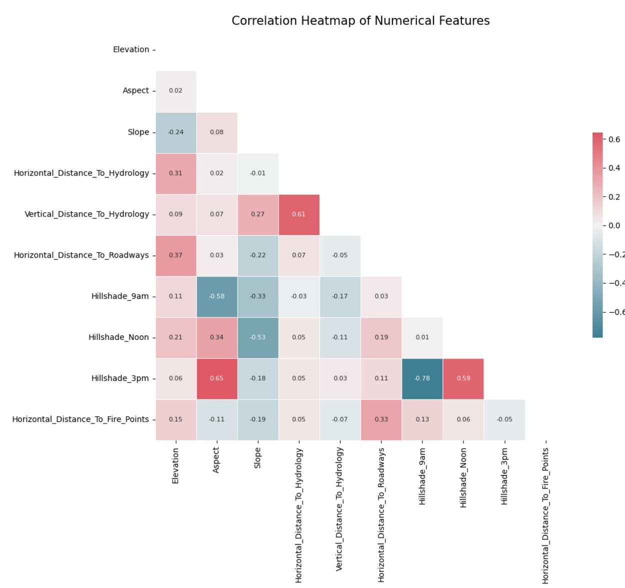

Figure 3: Correlation Heatmap of Numerical Features

Key observations from the correlation analysis:

- Horizontal distance to hydrology is highly correlated with vertical distance to hydrology (r 0.61), since streams in valleys mean that being far horizontally often also means being far vertically from water sources

- Hillshade at 9am is very negatively correlated with hillshade at 3pm (r -0.78), reflecting that east-facing slopes (bright in morning) are in shadow in late afternoon, and vice versa
- Hillshade at noon has a moderate negative correlation with slope (r -0.53), as steeper slopes receive less direct overhead sunlight
- Elevation shows a moderate correlation with horizontal distance to roadways (r 0.37), likely because roads are more common in accessible lower terrains

These correlations indicate some redundancy in the features and highlight underlying geophysical relationships in the data.

2.3.4 3D Visualization of Terrain Features

To better understand how cover types cluster in feature space, we plotted samples in a 3D space of Elevation, Slope, and Aspect with points colored by cover type.

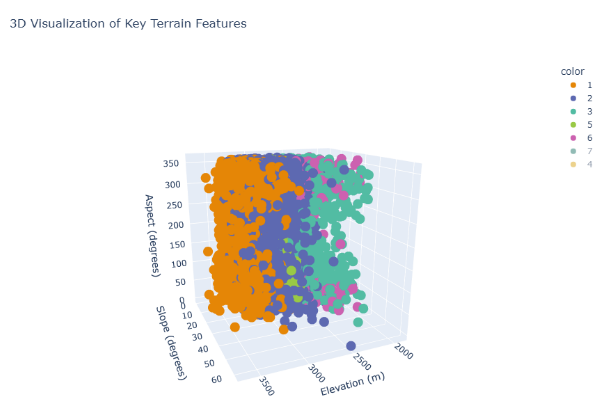

Figure 4: 3D Visualization of Key Terrain Features

This visualization (Figure 4) shows clear stratification along the elevation axis: the highest points were almost exclusively Class 7 (Krummholz), while the lowest elevation points belonged to Class 4 (Cottonwood/Willow). Classes 1 and 2 occupy broad mid-to-high elevation zones, overlapping each other but largely separated from lower-elevation classes. Class 3 (Ponderosa Pine) clusters toward the lower end of the elevation spectrum with moderate slopes. Classes 5 (Aspen) and 6 (Douglas-fir) lie in intermediate elevation bands and often in transitional areas.

When focusing on just the minority classes, we observed that while Class 7 is mostly distinct (very high elevation), Classes 5 and 6 overlap considerably with the lower end of Class 1 or the upper range of Class 3. This overlap foreshadowed the confusion the models would have in distinguishing those minority classes.

2.3.5 Terrain Feature Distributions

Examining the distribution of each terrain feature by forest cover type provides additional insights into the separability of classes (Figure 5).

These distributions reveal that beyond elevation, some cover types have unique profiles in other terrain features (slope, aspect, etc.), but significant overlap exists, necessitating additional features to reliably distinguish between overlapping classes.

3 Data Preprocessing

Before applying machine learning algorithms, several preprocessing steps were performed:

3.1 Data Splitting

The dataset was split into training, validation, and test sets with the following distribution:

- Training set: 348,607 samples (60%)
- Validation set: 116,202 samples (20%)
- Test set: 116,203 samples (20%)

The splitting was performed with stratification on the target variable to maintain the same class distribution across all subsets.

3.2 Feature Scaling

Numerical features were standardized using StandardScaler from scikit-learn, which transforms features to have zero mean and unit variance. This is particularly important for algorithms sensitive to feature scale, such as Logistic Regression.

$$z = \frac{x - \mu}{\sigma} \tag{1}$$

where μ is the mean of the feature values and σ is the standard deviation.

3.3 Feature Importance Analysis

A preliminary feature importance analysis using Random Forest was conducted to understand which features contribute most to the prediction (Figure 6).

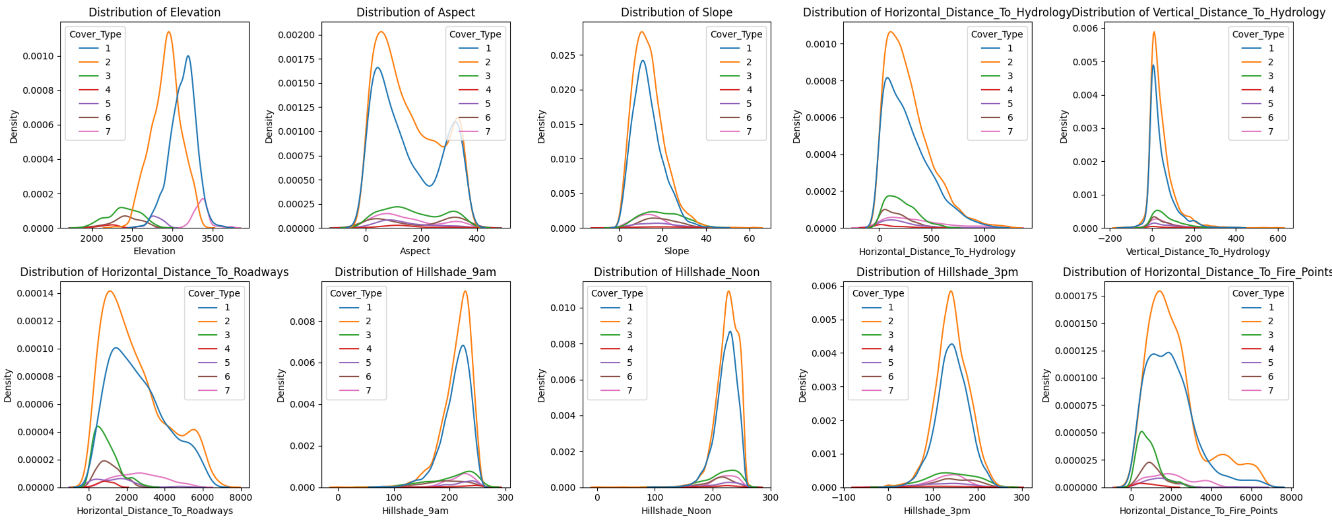

Figure 5: Distribution of Terrain Features by Forest Cover Type

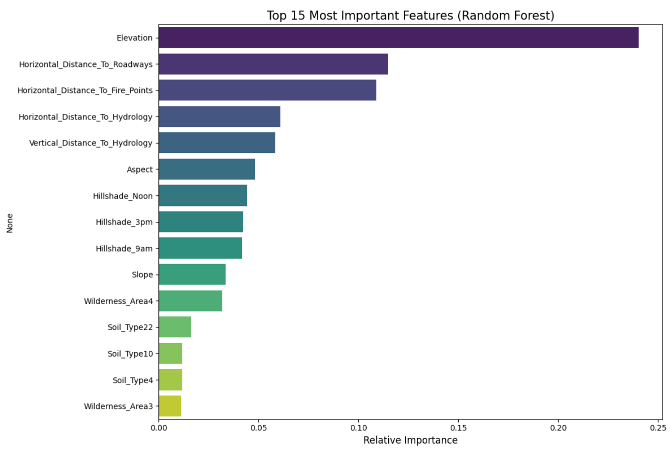

Figure 6: Top 15 Most Important Features Identified by Random Forest

The analysis revealed that elevation is the most important feature, followed by horizontal distance to roadways and horizontal distance to fire points. This information guided further analysis and model development.

4 Machine Learning Methods

We implemented three different machine learning algorithms for this classification task, each representing a different level of complexity and approach:

4.1 Logistic Regression

Logistic Regression is a fundamental classification algorithm that models the probability of class membership. For our multi-class problem, we used the one-vs-rest approach, training a separate logistic model for each of the seven forest cover types.

The logistic function (or sigmoid function) is given by:

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n)}} \tag{2}$$

where β_i are the model parameters and x_i are the feature values.

We implemented Logistic Regression using scikit-learn, with the 'saga' solver to handle the large dataset efficiently and L2 regularization to prevent overfitting. The model was run with a maximum of 4,000 iterations to ensure convergence.

Strengths:

- Interpretable model with clear feature coefficients
- Fast training even on large datasets
- Provides probability estimates for each class

Limitations:

- May not capture complex non-linear relationships in the data
- Assumes linear decision boundaries between classes
- Often struggles with imbalanced datasets

4.2 Random Forest

Random Forest is an ensemble method that builds multiple decision trees and merges their predictions. Each tree is trained on a random subset of the data and features (bagging), which helps reduce overfitting and improve generalization.

The prediction is made by aggregating the predictions of individual trees:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^{B} \hat{y}_b(x) \tag{3}$$

where B is the number of trees and $\hat{y}_b(x)$ is the prediction of the b-th tree.

Our implementation used 100 trees with no maximum depth limitation. This allowed the trees to capture complex patterns in the data while the ensemble nature of the model prevented overfitting.

Strengths:

- Handles complex non-linear relationships well
- Less prone to overfitting than single decision trees
- Can handle high-dimensional data with many features
- Provides reliable feature importance estimates
- Performs well on imbalanced datasets

Limitations:

- Less interpretable than linear models
- Can be computationally expensive for very large datasets
- May not be as memory-efficient as gradient boosting methods

4.3 LightGBM with Hyperparameter Tuning

LightGBM is an advanced gradient boosting framework that uses tree-based learning algorithms. Unlike Random Forest, which builds trees independently, gradient boosting builds trees sequentially, with each tree correcting the errors of the previous trees.

LightGBM employs a leaf-wise tree growth strategy rather than level-wise growth, which can lead to better accuracy with fewer trees. It's designed to be efficient and capable of handling large datasets with high dimensionality.

Hyperparameter tuning was performed using Optuna, an automatic hyperparameter optimization framework. We ran 20 trials, optimizing for the macro F1 score on the validation set. The key hyperparameters tuned were:

- num_leaves: Controls the complexity of the tree (range: 31-96)
- learning_rate: Step size shrinkage used to prevent overfitting (range: 0.03-0.2)
- feature_fraction: Fraction of features to consider for each tree (range: 0.75-1.0)
- bagging_fraction: Fraction of data to use for each tree (range: 0.75-1.0)

The optimal configuration found was: num_leaves=91, learning_rate0.087, feature_fraction0.932, and bagging_fraction0.841.

Strengths:

- Often achieves state-of-the-art performance
- Handles large datasets efficiently with lower memory usage
- Can capture complex patterns in the data
- Fast training and prediction speeds

Limitations:

- More complex to tune than simpler models
- Can overfit if not properly regularized
- Less intuitive feature importance calculation than Random Forest

5 Results and Discussion

5.1 Model Performance Comparison

We evaluated all three models on both validation and test sets using accuracy and macro F1 score. The macro F1 score was chosen as the primary metric due to the class imbalance, as it gives equal weight to each class regardless of frequency.

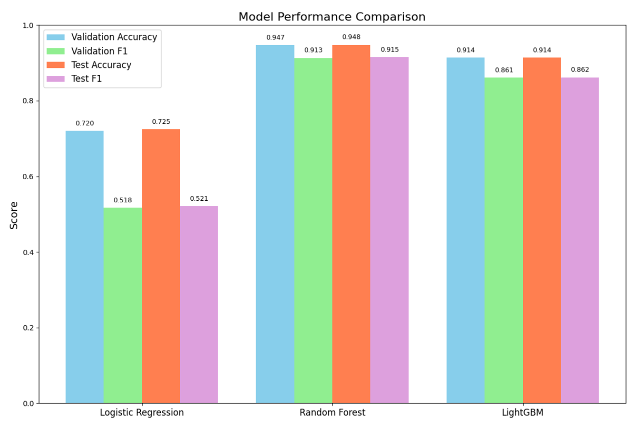

Figure 7: Performance Comparison of Machine Learning Models

Table 1: Validation and test performance for each model (scores from Fig. 7).

Model	Val. Accuracy	Val. F1	Test Acc.	Test F1
Logistic Regression	0.720	0.518	0.725	0.521
Random Forest	0.947	0.913	0.948	0.915
LightGBM	0.914	0.861	0.914	0.862

As shown in Figure 7, the ensemble methods dramatically outperformed the linear model:

- Logistic Regression: Test Accuracy 72.5%, Macro F1 0.52
- Random Forest: Test Accuracy 94.8%, Macro F1 0.915

- LightGBM: Test Accuracy 91.4%, Macro F1 0.862

The gap between accuracy and macro-F1 for Logistic Regression indicates it mostly predicted majority classes correctly but failed on minority classes. Both ensemble models had a small generalization gap (validation and test scores were similar), suggesting they did not overfit.

5.2 Confusion Matrices

Confusion matrices provide detailed insight into model performance across different classes.

5.2.1 Logistic Regression

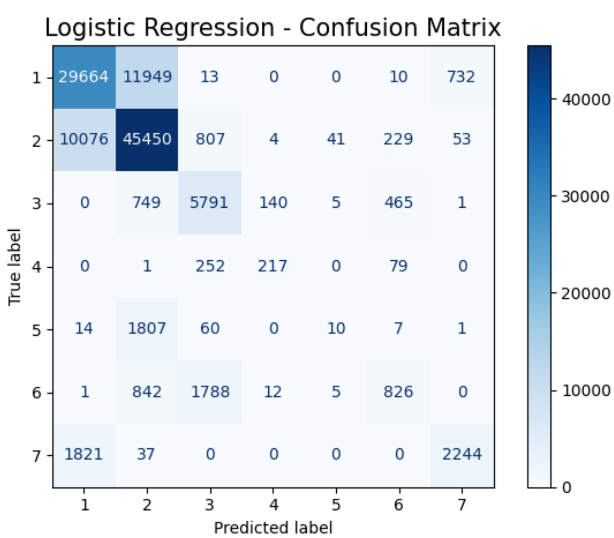

Figure 8: Confusion Matrix for Logistic Regression

The Logistic Regression model (Figure 8) shows poor performance on minority classes, particularly Class 5 (Aspen), where only 10 out of 1,899 samples were correctly classified. It performs acceptably on the majority classes (1 and 2).

5.2.2 Random Forest

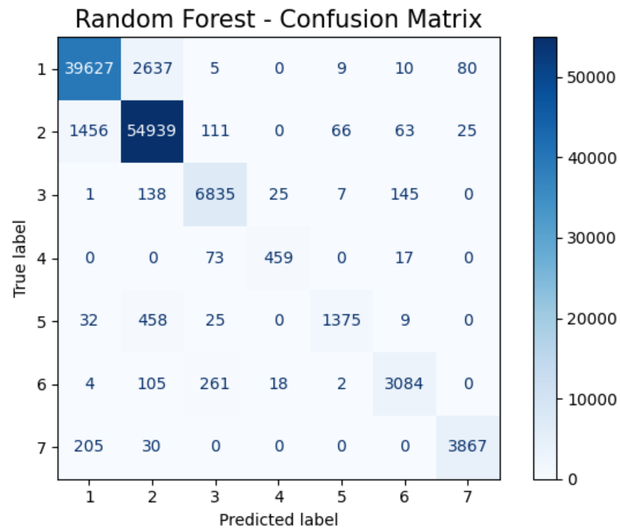

Figure 9: Confusion Matrix for Random Forest

The Random Forest model (Figure 9) demonstrates strong performance across all classes, including the minority classes. For Class 4 (Cottonwood/Willow), it achieved 84% recall (459 out of 549 samples correctly classified). Class 5 (Aspen) remained challenging but still saw good performance, with 1,375 out of 1,899 samples (72%) correctly classified.

5.2.3 LightGBM

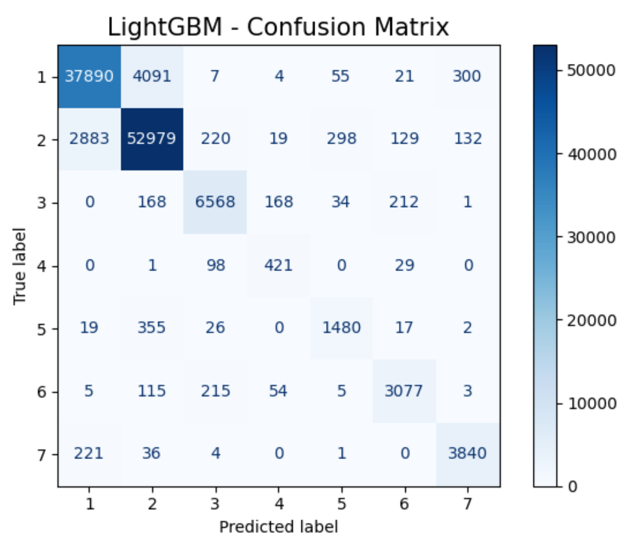

Figure 10: Confusion Matrix for LightGBM

The LightGBM model (Figure 10) shows good performance across classes, though slightly lower than Random Forest on some minority classes. It correctly classified 1,480 out of 1,899 Aspen samples (78%) and 421 out of 549 Cottonwood/Willow samples (77%).

5.3 Per-Class Performance

Per-class F1 scores provide a more granular understanding of model behavior.

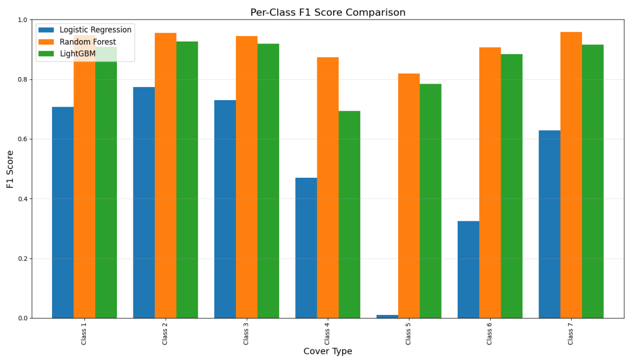

Figure 11: Per-Class F1 Score Comparison

Figure 11 shows stark differences in model performance by class:

- For the majority classes (1 and 2), Logistic Regression achieved F1 scores of approximately 0.71 and 0.77, respectively
- For minority classes, Logistic Regression performed poorly, with F1 scores below 0.5, and for Class 5 (Aspen), the F1 score was near zero (0.01)
- Random Forest maintained F1 scores above 0.82 for all classes, even the rare ones (F1 0.87 for Class 4, 0.82 for Class 5)
- LightGBM's per-class F1 scores were slightly lower than Random Forest's on the minority classes but still strong (typically in the 0.75-0.80 range)
- Both ensemble models performed exceptionally well on Classes 1, 2, and 7, with F1 scores around 0.95

Class 7 (Krummholz) is a minority class (3.5% of data) but was easier to distinguish due to its very high elevation range, allowing all models to perform relatively well on it.

5.4 Feature Importance

5.4.1 Random Forest Feature Importance

The feature importance analysis from Random Forest (Figure 6) identified elevation as the most important feature, followed by horizontal distance to roadways and horizontal distance to fire points.

5.4.2 LightGBM Feature Importance

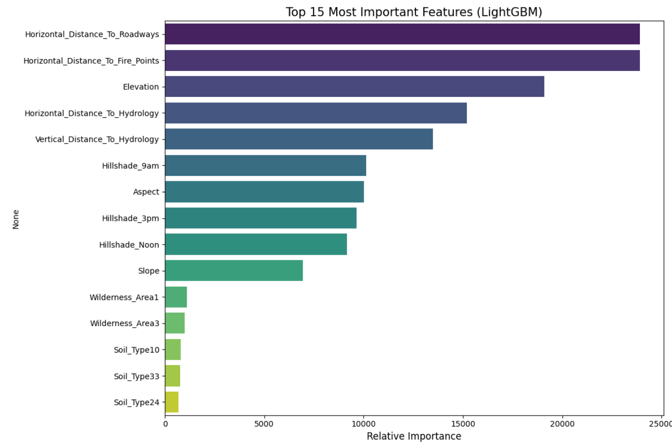

Figure 12: Feature Importance from LightGBM

The LightGBM feature importance (Figure 12) shows that horizontal distance to roadways and horizontal distance to fire points are the most important features, followed by elevation. This slight difference from the Random Forest importance ranking could be due to the different algorithms used to calculate feature importance.

5.5 Misclassification Analysis

To identify patterns in model errors, we conducted a misclassification analysis by visualizing test samples in the 3D terrain feature space, with colors indicating how many models misclassified each point.

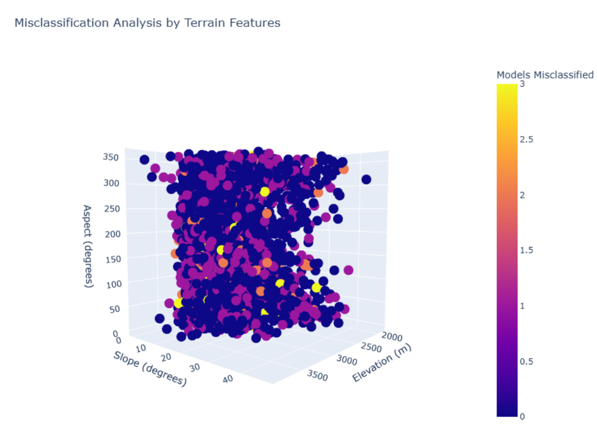

Figure 13: Misclassification Analysis by Terrain Features

The visualization (Figure 13) revealed that misclassifications are not randomly distributed. Most challenging cases (points misclassified by all models) form clusters in specific regions of the terrain feature space. Key findings include:

- Many consistently misclassified points appear at boundaries between cover type clusters or in areas with ambiguous feature values

- Numerous misclassifications occur at moderate elevations (2,800–3,000 m) where Spruce/Fir transitions to Alpine tundra or where Lodgepole Pine transitions to Ponderosa Pine
- Almost all points from the rare Class 4 (Cottonwood/Willow) were misclassified by Logistic Regression, though the tree-based models correctly identified many of them
- Class 5 (Aspen) proved especially challenging since Aspen stands often occur intermixed with other forest types at similar elevations and slopes
- Points mispredicted by all models typically represent truly ambiguous cases given the available features

The misclassification patterns highlight a stark contrast between the logistic model and the ensembles. Logistic Regression misclassified many points (especially from minority classes) that were correctly handled by Random Forest and LightGBM, suggesting these points lie in regions where the relationship between features and class is non-linear or involves complex interactions.

These insights suggest that most remaining errors correspond to ecological transition zones and overlapping forest types in feature space, with minority classes being particularly challenging due to limited examples and inherent feature overlap.

6 Conclusion

This project demonstrated the application of machine learning techniques to predict forest cover types based on cartographic variables. Several important conclusions can be drawn:

6.1 Model Performance

The ensemble methods (Random Forest and LightGBM) significantly outperformed the simpler Logistic Regression model, with Random Forest achieving the highest macro F1 score of 0.915. This confirms the superiority of ensemble methods for complex ecological classification problems with class imbalance.

Random Forest was the best overall performer, with approximately 94.8% accuracy and 0.915 macro-F1, slightly outperforming LightGBM's 91.4% accuracy and 0.862 macro-F1. The difference was most pronounced on minority classes. Logistic Regression's performance was inadequate for practical application, with just 52% macro-F1 despite 72% accuracy.

6.2 Feature Importance

Terrain features, particularly elevation and distances to infrastructure (roads and fire points), were consistently identified as the most important predictors of forest cover type. In the Random Forest model, Elevation accounted for roughly 23-25% of total importance—more than 1.5× higher than any other feature. This aligns with ecological knowledge about the influence of elevation gradients and human activities on forest composition.

The fact that distances to roads and fire points ranked highly suggests that human access and disturbance patterns significantly influence forest composition. Microclimate factors like hillshade and aspect also played important secondary roles, while soil type indicators were surprisingly less influential despite their ecological relevance.

6.3 Class Imbalance Challenges

The class imbalance in the dataset presented significant challenges. Logistic Regression was essentially incapable of correctly identifying minority classes like Class 5 (Aspen), where it achieved an F1 score of only 0.01. The ensemble methods were far better equipped to handle this imbalance, though even they showed lower performance on minority classes compared to majority classes.

This underscores the importance of using appropriate evaluation metrics (like macro-F1 rather than just accuracy) and choosing algorithms that can effectively learn from limited examples of minority classes.

6.4 Ecological Insights

The analysis revealed distinct patterns in how different forest types are distributed across the landscape:

- Class 7 (Krummholz) is found almost exclusively at very high elevations (above 3,300 m)
- Class 4 (Cottonwood/Willow) occurs in low-elevation riparian areas
- Classes 1 (Spruce/Fir) and 2 (Lodgepole Pine) dominate mid-to-high elevations
- Class 5 (Aspen) and Class 6 (Douglas-fir) often occur in transitional zones

The misclassification analysis suggested that many errors occur at ecological transition zones, particularly at moderate elevations where multiple forest types can potentially exist. These transition zones represent natural ecological complexity rather than model limitations.

6.5 Personal Insights and Learning Outcomes

Throughout this project, I gained valuable insights into both machine learning techniques and ecological modeling:

- I learned how to effectively handle large datasets (580,000+ samples) with significant class imbalance
- I developed a deeper understanding of ensemble methods and their advantages over simpler models
- I improved my skills in hyperparameter tuning using automated frameworks like Optuna

- I gained appreciation for the importance of thorough exploratory data analysis before model building
- I discovered how machine learning can provide valuable insights into complex ecological systems
- I learned the importance of choosing appropriate evaluation metrics, especially in imbalanced classification problems
- I developed a better understanding of the relationship between geographic features and forest ecosystems

6.6 Practical Applications

The models and insights developed in this project have several practical applications:

- They can help forest managers make more informed decisions about resource allocation
- They can aid in monitoring changes in forest ecosystems over time
- They can potentially help predict how climate change might affect forest distributions
- They demonstrate how machine learning can contribute to environmental research and management

6.7 Future Work

Several avenues for future improvement and extension of this work exist:

- Further hyperparameter tuning with more extensive Optuna trials
- Feature engineering to better capture terrain relationships, particularly in transition zones
- Addressing class imbalance using techniques like SMOTE to improve performance on minority classes
- Exploring other ensemble methods like XGBoost or Stacking Classifiers

7 Acknowledgements

I acknowledge the use of several large language models (LLMs), including Claude, Gemini, and ChatGPT, to assist with brainstorming, code debugging, optimization suggestions, project structuring, and writing this report. These AI assistants helped me refine my implementation of machine learning algorithms and visualization techniques. The core implementation and understanding of the algorithms used in this project reflect my personal learning journey through the course materials. The project dataset was obtained from the UCI Machine Learning Repository, and I thank the original contributors for making this data publicly available for educational purposes.

8 References

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9 Source Code

The complete source code for this project is available on GitHub: https://github.com/MonishSoundarRaj/forest-cover-classification